

Akshay Bapat

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

Master of Science in Computer Science, Concentration in Artificial Intelligence

University of Illinois, Urbana – Champaign

Bachelor of Science in Computer Engineering

WORK EXPERIENCE

Invisible Technologies

NYC, NY

Forward Deployed Engineer @ Apollo Global Management

2026–Present

- **Designed and shipped an MCP authorization gateway** that routes AI agents to downstream tool servers (Workday, ServiceNow, HR knowledge, compliance) with Okta JWT validation, tool-scope filtering, Auth elicitation flows, and structured audit logs capturing tool inputs/outputs and user identity. Took it from design to production in 6 weeks
- **Delivered a production RAG compliance chatbot** (Azure AI Search + GPT-4o, FastAPI, Next.js) with **2,300+ monthly users** from auth and mobile-ready UI. Exposed it to agents via MCP and a LRTC gateway so Claude/Cursor-style clients can query grounded, cited policy answers through the gateway with **99% accuracy measured by a custom eval harness**
- Established LLM observability as a practice at Apollo pitched and demoed to the Ai Engineering Org. Then **authored a shared observability library (corptech-ai-observability)** adopted across 6+ services, standardizing Datadog LLM Observability, OpenTelemetry tracing, and PII redaction.
- Currently building a **remediation agent** that detects latency spikes, bad retrievals, and unhealthy usage signals, proposes fixes, and executes approved remediations through the MCP gateway with human-approval gates and full audit trails, closing the loop from observability to action.
- Partner directly with Apollo stakeholders across compliance, HR, and engineering to scope deployments, run demos, and drive adoption as the embedded technical owner of the AI agent platform.

Northern Trust

Chicago, IL

AI Developer

2023-2026

- Architected AI system using distributed agents, **Python, and LLMs(GPT-4, LangChain, Autogen)** that automated legacy code migration, reducing manual conversion time by 85% across 500K+ lines of code and saving \$1.2M in engineering costs using evaluation metrics, writing custom evals to measure accuracy and reliability, and prompt tuning to improve model outputs by 15%.
- Built enterprise ML-driven classification pipeline for SNOW incident tickets using Azure Machine Learning and Cognitive Search with (RAG) framework automating ticket categorization, reduced triage/resolution time, and enabled natural language Q&A providing support staff with resolution steps and client inquiry responses.
- Orchestrated cloud infrastructure on **Azure using Kubernetes, Terraform, and Docker**, reducing cloud costs while improving system reliability and scalability

INDEPENDENT TECHNICAL PROJECTS

Scientific Paper Retrieval-Augmented Generation System | Python, FAISS, Transformers, Langchain

- Designed an embedding and retrieval pipeline for research papers using transformer models and vector search.
- Implemented context-aware question answering and text summarization, improving retrieval relevance by 35% over baseline keyword search.
- Integrated dataset evaluation metrics and prompt-tuning loops for response quality analysis.

GenAI-powered Multi-Agent Sales CRM | Firebase, Grok, MCP, Tensorflow

- Built conversational AI agents leveraging LLMs for dialogue generation, intent classification, and personalized outreach.
- Designed language-data experiments to evaluate dialog coherence and user engagement metrics.